# zpc60 *sampling sequencer*



## The keys




| | |
|---|---|
| **White 1–16** | the **16 pads** of bank **A** (or **B**): white 1 = pad 01, white 16 = pad 16 |
| **OCT+** **OCT–** | bank **B** / back to bank **A** (the LEDs A B show which) |
| **black keys** | **loops 1–11** (12–16: SEQ → LOOP): stopped at once; playing: from the end of the loop (its key blinks, the LCD says **LOOP 3 NEXT**) |
| **ENV held + pads** | **NOTE REPEAT**: the pads held hit on every step of AUTO CORRECT (OFF: 1/16), swung; recording, they go in. Stopped: at once, then in tempo. ENV tapped: tap tempo |
| **ARP** | PADS → **MULTI PITCH** (the last sound on all 27 keys, F4 as tuned) → **16 LEVELS** (white 1–16: the last sound, soft to loud) → PADS |
| **MIDI in** | notes 36–67 play A01–B16 (velocity in 16 steps); Program Change 0–15 picks loop 1–16 |

## The factory pads

**BANK A**

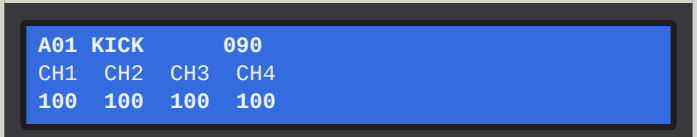
| | | | |
|---|---|---|---|
| **13** CONGA | **14** RIDE | **15** SCRCH | **16** COWBL |
| **09** TOM L | **10** RIM | **11** OHAT | **12** SNAP |
| **05** TOM H | **06** CRASH | **07** BASS | **08** EP Dm9 |
| **01** KICK | **02** SNARE | **03** HAT | **04** CLAP |

**BANK B**

| | | | |
|---|---|---|---|
| **13** WOOD | **14** CONGA +4 | **15** SCRCH −5 | **16** SNAP 33 |
| **09** TOM L −5 | **10** RIM 33 | **11** SHAKR | **12** TAMB |
| **05** CLAVE | **06** CONGA −5 | **07** BASS −5 | **08** EP −7 (Gm9) |
| **01** KICK 33 | **02** SNARE 33 | **03** HAT 33 | **04** CLAP 33 |

Pad 01 bottom left, as on the old sampling sequencers. **33**: TUNE −7 + SPEED 33, darker and grainier. Any pad, any of the 20 samples or your own: hit it, turn **PRESETS**. 12 bit at 40 kHz, pitched without interpolation.

## The knobs




| | |
|---|---|
| **KNOB 1–4** | the **faders** of channels 1–4 (100 as set, 50 −12 dB); **SEL**: **5–8** (lit) / 1–4. The LCD's four columns always say what the knobs do |
| **A page** | EDIT, FX, SEQ, GLO open one (tap again: the next); untouched **12 s** (SETUP: BACK), or **HOME**: the pads and faders again |
| **SELECT** | tempo |
| **ALGORITHM** | the pad to edit (A01 … B16), without playing it |
| **PRESETS** | **the sample** of the selected pad: one sample on many pads, each with its own TUNE, DECAY … |
| **MASTER** | the volume |

## Pages · KNOB 1 · 2 · 3 · 4

| | |
|---|---|
| **EDIT WAVE** | the pad's sample · COPY> pad · COPY (turn twice) · – |
| **SOUND** | TUNE (semitones) · FINE · DECAY · LEVEL |
| **TRUNC** | START · END · DIR (fwd / rev) · SPEED (45 / 33) |
| **OUT** | CHOP group (--, 1–8: its pads cut each other) · PAN · CUT · RESO (only channels 1–2 filter) |
| **SENDS** | DRIVE · CHO · DLY · REV: **the effects of this pad** |
| **FX FILTER** | **DJ filter** on the mix: left low-pass, right high-pass, centre OFF · RESO |
| **CHORUS** | RATE · DEPTH · MIX |
| **DELAY** | TIME (1/4 … 16T, in tempo) · FDBK · COLOR · MIX |
| **REVERB** | SIZE · DAMP · PRE (ms) |
| **SEQ LOOP** | LOOP · BARS (AUTO, 1–32) · QUANT (AUTO CORRECT, also NOTE REPEAT's grid) · SWING 50–71 % |
| **LOOP TOOLS** | CLEAR (turn twice) · COPY> loop · COPY (turn twice) |
| **SONG** | STEP · SEG (its loop) · REPEAT (0: the song ends here) · SONG OFF / 1–4 |
| **GLO SETUP** | TEMPO · BACK · – · RESET (**all loops gone**, turn twice); tap: CLICK: CLICK · COUNT · DUB; tap: OUTPUT: SMOOTH |

## Make a beat

1. **REC** (**ARMED**), **PLAY**: **COUNT 4 3 2 1**, then it records.
2. Kick (white 1) and snare (white 2). **AUTO CORRECT** puts them on the grid.
3. **REC** again at the end: an empty loop (BARS **AUTO**) is now as long as your take.
4. Hats with **NOTE REPEAT**: **REC**, hold **ENV** and white 3: 1/16 hats while you hold. QUANT 1/8 or 1/32 for other rolls, SWING for the feel.
5. Or step by step: hit the hat, **hold** **SEQ**: the white keys are its 16 steps; **OCT+** / **OCT–** the next / last bar.
6. Ghost notes: **ARP** twice (**16 LEVELS**), white 1–16 play the snare soft to loud.
7. Another loop: a black key, record into it. Songs: SEQ → SONG.

## Buttons

| | |
|---|---|
| **PLAY** | run / stop |
| **REC** | stopped: ARMED; playing: overdub on / off; **hold 2 s**: clear the loop |
| **ENV** | tapped: tap tempo · held + pads: **NOTE REPEAT** |
| **HOME** | from a page: the pads · on the pads: **TRACK MUTE** on / off (HOME blinks): white keys mute / unmute their pad, **ENV** held + pad: **solo**; muted pads stay muted when you leave |
| **GLO held + white** | 1–8 **mute**, 9–16 **solo** ch 1–8 |
| **EDIT held + OCT–** | **undo** the last clear / erase / copy (again: redo) |
| **FX held + white 1–4** | **punch-in**: roll 1/8 · roll 1/16 · reverse · tape stop; **OCT–** / **OCT+**: reverb / echo throw |
| **LFO + pad** | **erase** its hits as the playhead passes (stopped: all) |
| **SAVE** | save now; **held + a black key**: the loop into that loop |

## Channels

8 channels, one sound each: a pad plays on **its place in the row pair**: pads 01 and 09 on channel 1, 02 and 10 on 2 … 08 and 16 on 8 (in both banks). **A new hit on a channel cuts the last**: HAT (03) and OHAT (11) choke. Channels 1–2 (pads 01 02 09 10) have the resonant filter, 3–6 a fixed low-pass, 7–8 none.

Based on zp12, SLOOP (isod89) and Felucca (Leo Kuroshita, Hügelton Instruments), GPL-3.0 · sounds CC0 (VSCO-2 CE, Sonic Pi), E-piano Greg Sullivan (CC BY 3.0)

Inspired by the sampling sequencers of the late 80s, names their owners' · not affiliated with M-VAVE or Akai

## What the lights say

| | |
|---|---|
| PLAY | lit while it runs |
| REC | lit: recording · blinking: ARMED, or an overdub waiting for the next 1 |
| ENV | blinks on the beat (tap along) |
| ARP · SEL | MULTI PITCH or 16 LEVELS on · the knobs are faders 5–8 |
| LFO | erasing (held + a pad) |
| SAVE | lit in song mode |
| OCT− OCT+ | bank A / B on the keys |
| HOME | blinking: TRACK MUTE; the white keys lit while their pad is heard, glowing when muted |
| **White keys** | flash when their pad sounds; **SEQ held**: the steps with a hit; **GLO held**: 1–8 the channels you hear, 9–16 the soloed |
| **Black keys** | the loop playing lit, the next one blinking |

## The screen

| | |
|---|---|
| **Big line** | the selected pad and its sound (A01 KICK), or what just changed |
| **Top right** | the tempo, with **ARMED**, **COUNT** or **REC** before it; below it the page |
| **4 × 4 pads** | the bank on the keys, pad 01 bottom left; red while it sounds, the selected one outlined, **dark: muted**, yellow ring: soloed |
| **Faders** | beside the pads: channels 1–8, the four on the knobs white; M muted, S soloed |
| **A B** | the bank on the keys |
| **M · L** | MULTI PITCH · 16 LEVELS |
| **S · NR** | song mode · NOTE REPEAT running |
| **RUN · REC** | running (blinks on the beat) · recording (blinks: ARMED) |
| **Red dot** | in the step grid: a hit with its own TUNE / FINE / DECAY / CUT |

## Recording

| | |
|---|---|
| **Count-in** | GLO → CLICK → COUNT: OFF, 1 bar (4 3 2 1), 2 bars |
| **AUTO** | a loop with BARS AUTO takes the length of its first take, in whole bars |
| **QUANT** | AUTO CORRECT: OFF, 1/8, 1/8T, **1/16**, 1/16T, 1/32, 1/32T |
| **SWING** | 50, 54, 58, 63, 67, 71 %: the off 1/16s (QUANT 1/8: the off 1/8s) come later; NOTE REPEAT swings along |
| **NOTE REPEAT** | its grid is QUANT (OFF: 1/16); playing, the first hit waits for the grid; every repeat is recorded like a hit |
| **Velocity** | 16 steps. The keys play at one level (14 of 16), **16 LEVELS** gives all 16, MIDI in its own |
| **Overdub** | a hit where one is: the louder stays. Up to 512 hits a loop |
| **Per hit** | TUNE, FINE, DECAY, CUT turned while recording stay with the pad's hits |
| **Step grid** | 16 steps of 1/16 a bar; a step set by hand plays at level 14 |

## Loops and songs

| | |
|---|---|
| **16 loops** | 1–11 on the black keys, 12–16 with SEQ → LOOP; each 1–32 bars or AUTO |
| **Copy** | SAVE held + a black key, or SEQ → TOOLS |
| **Songs** | 4 songs of up to 32 steps, each a loop (SEG) played REPEAT times; PLAY plays it from step 1 |

## Ranges

| | |
|---|---|
| **TEMPO** | 40–240 BPM; SELECT 0.5 a click, ENV tapped |
| **Faders** | 0–127: 100 as set, 50 −12 dB |
| **TUNE · FINE** | −24 … +12 semitones · ±50 cents |
| **Samples** | the kit 12 bit at 40 kHz; your own at 40, 26 or 20 kHz (less rate, more time) |
| **Filters** | channels 1–2: CUT, RESO (4-pole low-pass) · 3–6: fixed low-pass · 7–8: none |

## Saving and safety

| | |
|---|---|
| **Saved** | by itself when stopped and 3 s after a change; SAVE now |
| **Its own** | zpc60 keeps its own save; it never reads zp12's (they share the flash: switching back and forth, the last one saving wins) |
| **Not saved** | TRACK MUTE, mute / solo and the DJ filter |
| **Undo** | EDIT held + OCT−: the last clear, copy or stopped erase |
| **RESET** | GLO → SETUP: everything back to the factory, **all loops gone** |
| **Rescue** | no answer? Hold OCT− while switching on: the USB rescue, then install again |



GPL-3.0 · github.com/zvenson/zpc60